

Online Experiments for Language Scientists

Lecture 6: Perceptual learning

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Adapted from slides by Kenny Smith

Today's paper

Lev-Ari, S. (2017). Talking to fewer people leads to having more malleable linguistic representations. *PLoS ONE*, 12, e0183593.

Perceptual learning experiment run on MTurk

- Do people with smaller social networks have more malleable linguistic representations (as assessed through a perceptual learning experiment)?



Shiri Lev-Ari
(Royal Holloway,
University of
London)

Rationale

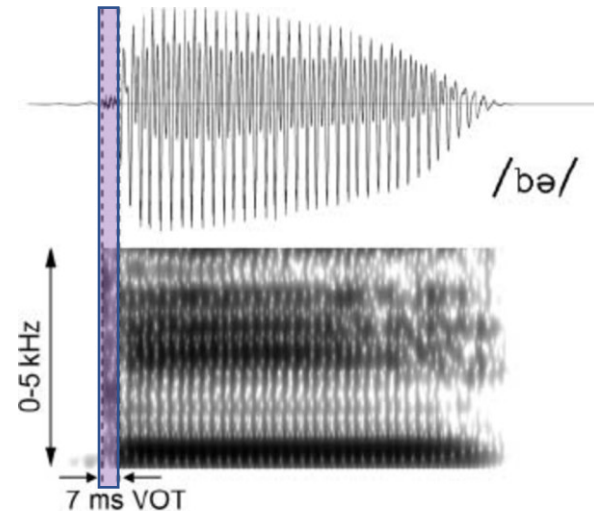


Rationale

- Inverse relationship between how informative each source is and how many sources we already have
- Applied to **linguistic input**:
Exposure to relatively few speakers makes each new speaker more influential

Voice Onset Time

voiced stops → short VOT



voiceless stops → long VOT

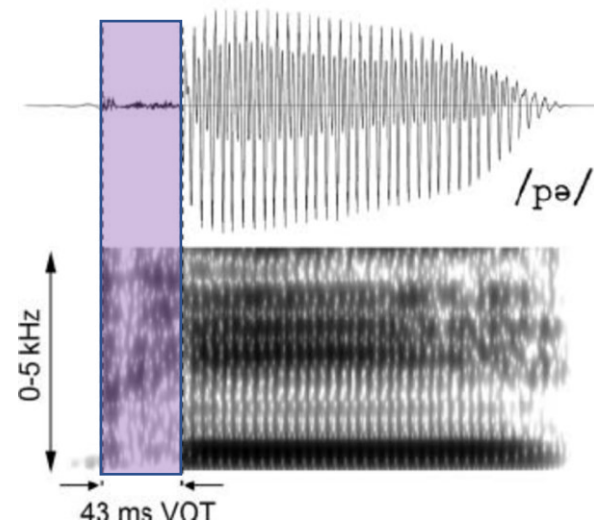


Image from Neef, N. et al. (2012). Reduced Speech Perceptual Acuity for Stop Consonants in Individuals Who Stutter. *Journal of Speech, Language, and Hearing Research*, 55, 276-89.

Voice Onset Time and /d/-/t/



5ms

Voice Onset Time and /d/-/t/



5ms



10ms

Voice Onset Time and /d/-/t/



5ms



10ms



15ms

Voice Onset Time and /d/-/t/



5ms



10ms



15ms



20ms

Voice Onset Time and /d/-/t/



5ms



10ms



15ms



20ms



25ms

Voice Onset Time and /d/-/t/



5ms



10ms



15ms



20ms



25ms



30ms

Voice Onset Time and /d/-/t/



5ms



10ms



15ms



20ms



25ms



30ms



35ms

Voice Onset Time and /d/-/t/



5ms



10ms



15ms



20ms



25ms



30ms



35ms



50ms

Voice Onset Time and /d/-/t/



5ms



10ms



15ms



20ms



25ms



30ms



35ms



50ms



80ms

Perceptual learning task

- Intermediate VOT tokens + /d/ or /t/ interpretations shift categorical boundaries



25ms

Manipulated audio presentation



Unmanipulated



Manipulated (VOT=24ms)



Unmanipulated



Manipulated (VOT=24ms)



Perceptual learning task

- Intermediate VOT tokens + /d/ or /t/ interpretations shift categorical boundaries



25ms

- Does social network size modulate this effect?
 - Larger effect predicted for people with smaller social network

Measuring social network size

- Not experimentally manipulated
- Estimating people's actual social network size
 - Social network questionnaire

In this questionnaire we would like to gather information about your linguistic interactions. We realize that some of the estimates are difficult to make. Please do your best and be as accurate as possible.

Important: When providing estimates for your exposure in a week, keep in mind that your habits may vary considerably depending on the day of the week (e.g., weekday vs. weekend). Please be as accurate as possible and do not simply multiply your estimate for one day by 7.

1) How old are you?

2) With how many people do you converse orally **in a typical week**? (Please only include people with whom you regularly talk for longer than 5 minutes)

3) How many hours do you usually spend on conversing orally with people **in a typical week**?

Manipulated audio presentation

- Manipulated /d/ condition:
 - manipulated /d/ recordings and unmanipulated /t/ recordings
 - fillers
- Manipulated /t/ condition:
 - manipulated /t/ recordings and unmanipulated /d/ recordings
 - fillers

Design

- VOT manipulation: manipulated /d/ vs. manipulated /t/
- Test type: new speaker vs. same speaker (control)

Predicted to be affected by social network size:

	new speaker	same speaker
manipulated /d/		
manipulated /t/		

Predictions

Social network size:

- should ***not*** affect **learning** of phonological categories
- should affect **generalization** of phonological categories

Procedure

Social network questionnaire



Picture selection task (manipulated /d/ or manipulated /t/)



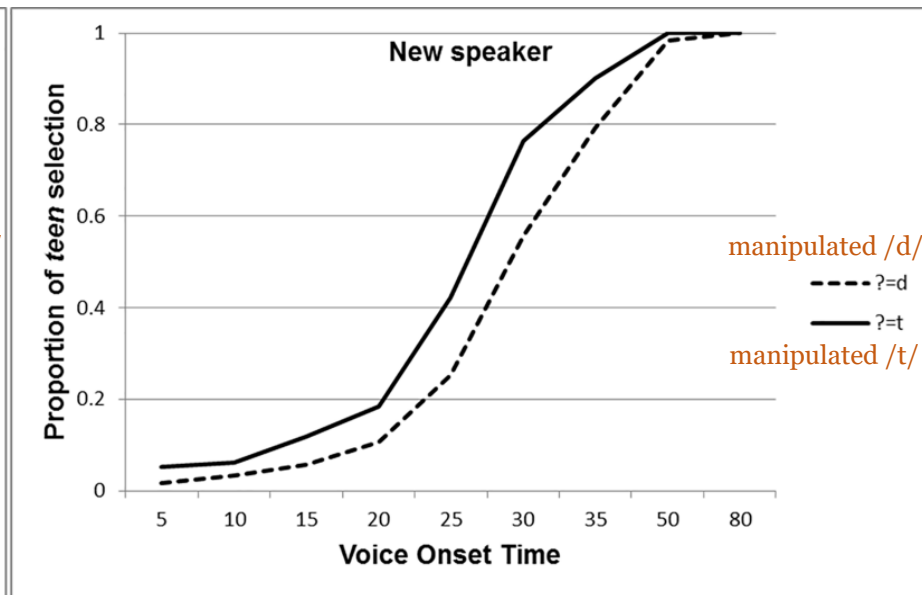
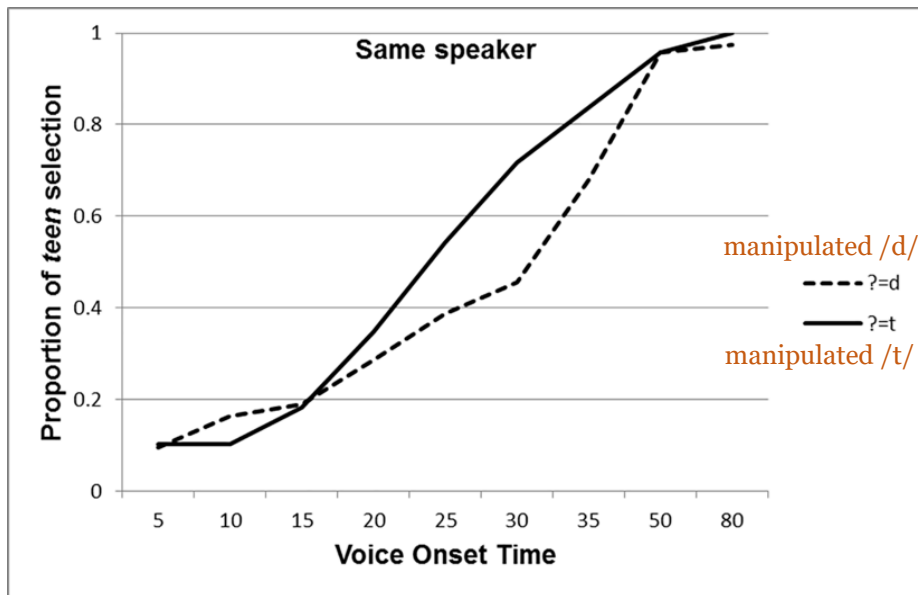
Phoneme categorization task (same or different speaker)

Demo using this week's practical code

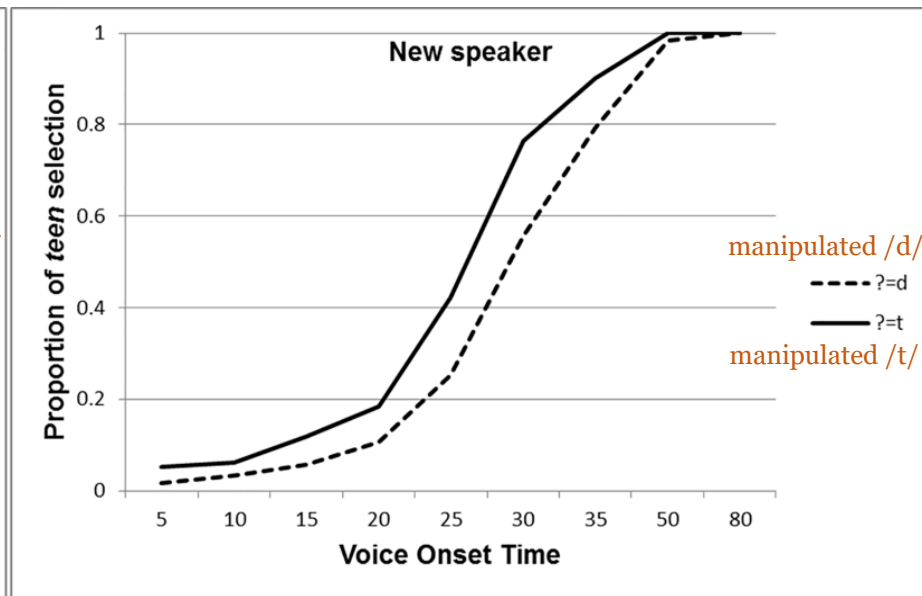
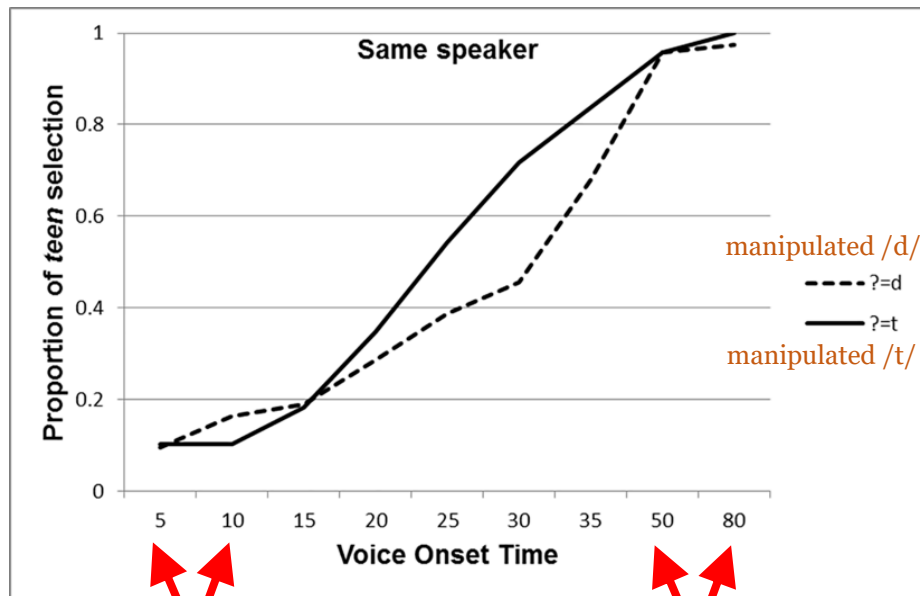
Sample size, study duration etc

- US-based MTurk workers
- N=148 after 5 exclusions (spread over 4 conditions)
- 5-15 minutes, \$1.20

Perceptual learning



Perceptual learning

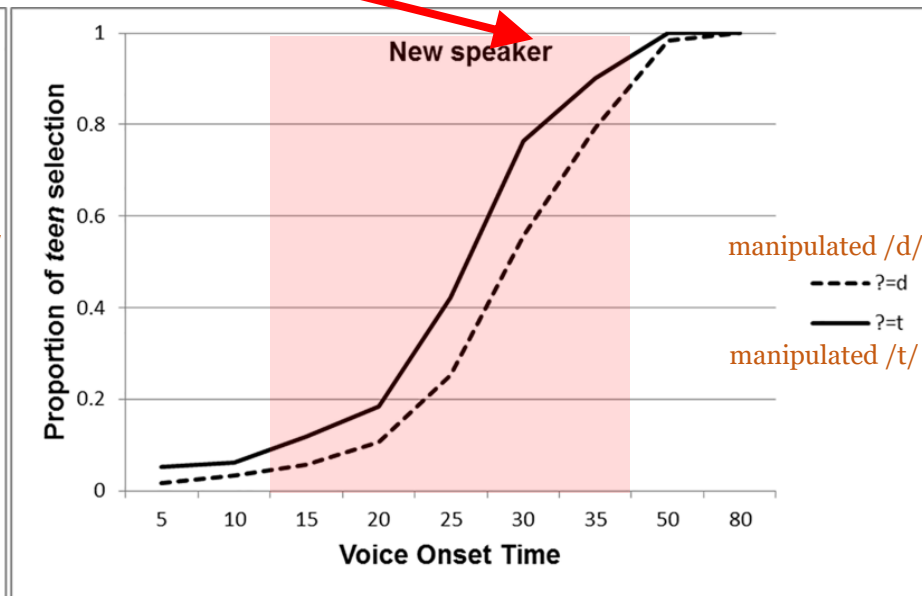
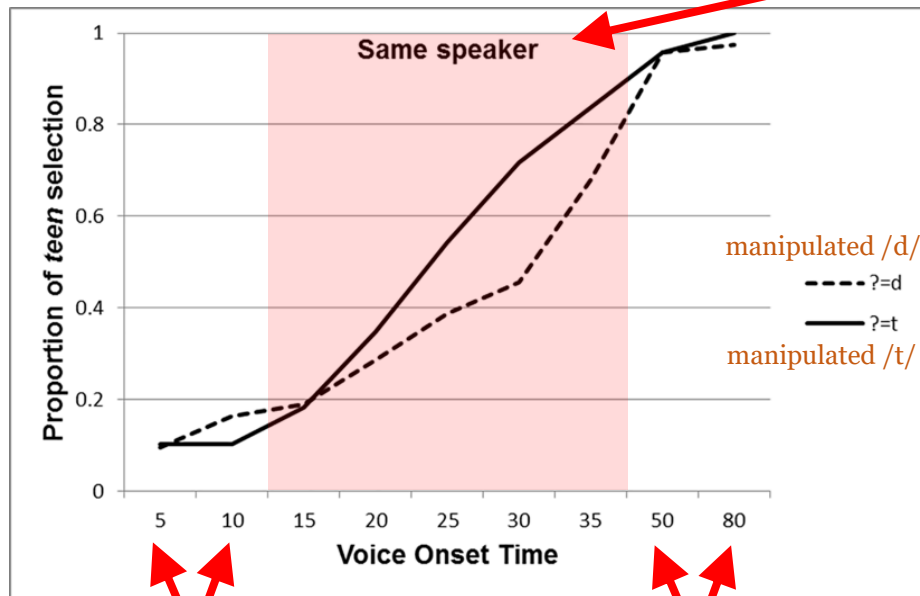


Good /d/

Good /t/

Perceptual learning

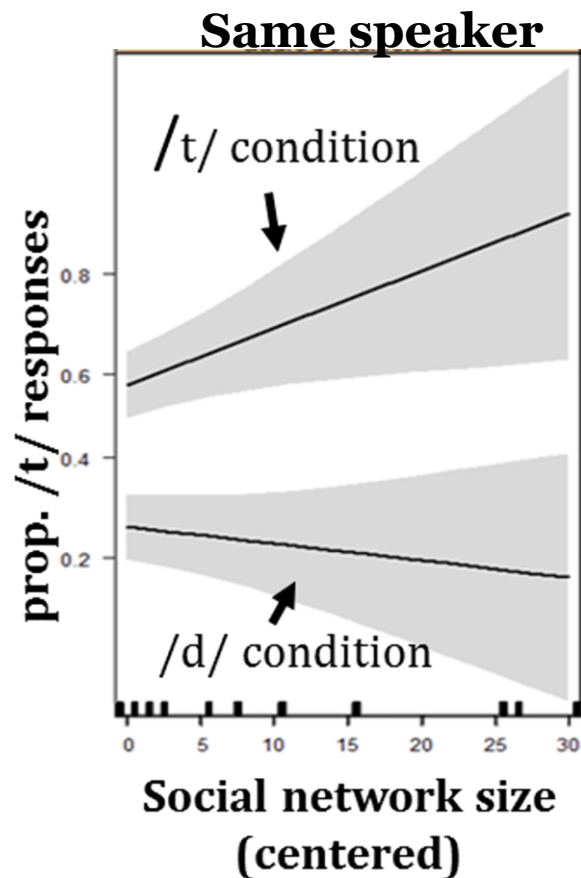
Ambiguous zone



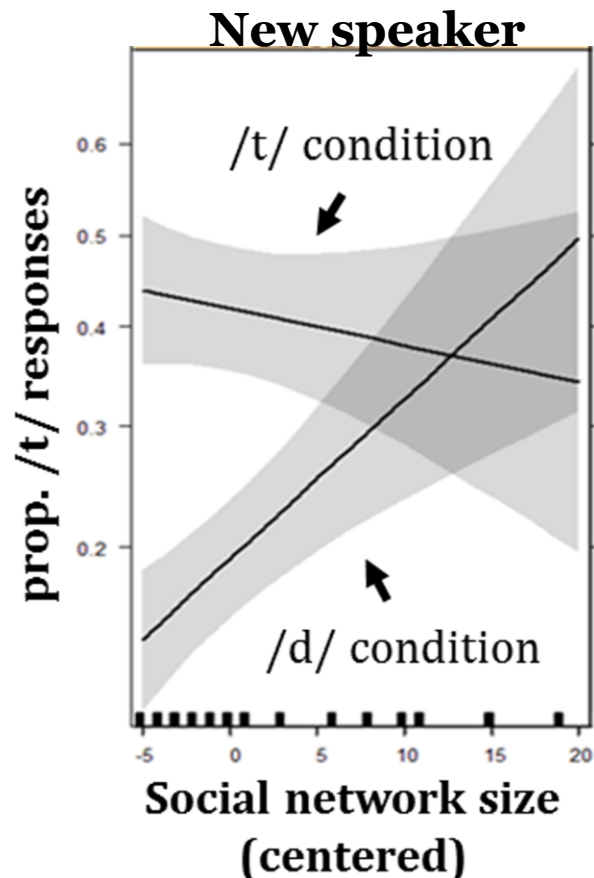
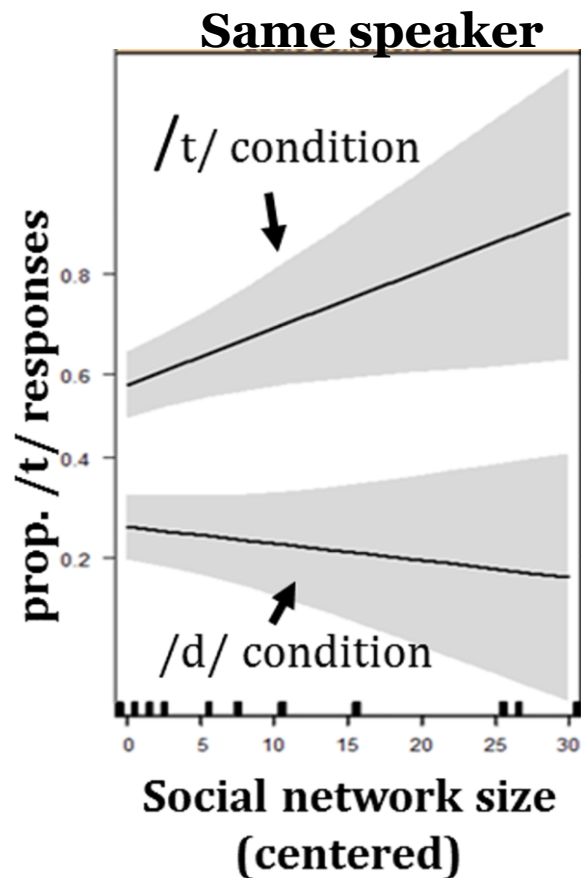
Good /d/

Good /t/

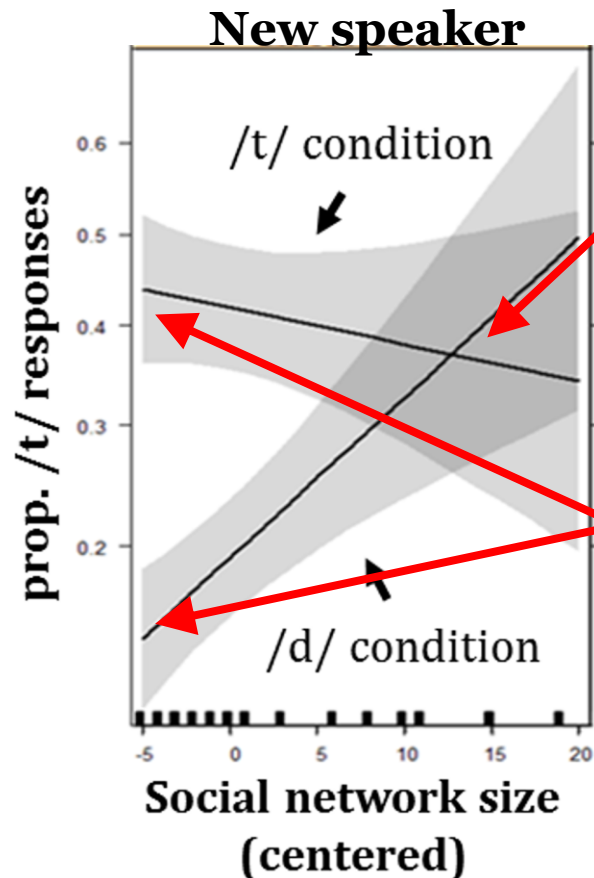
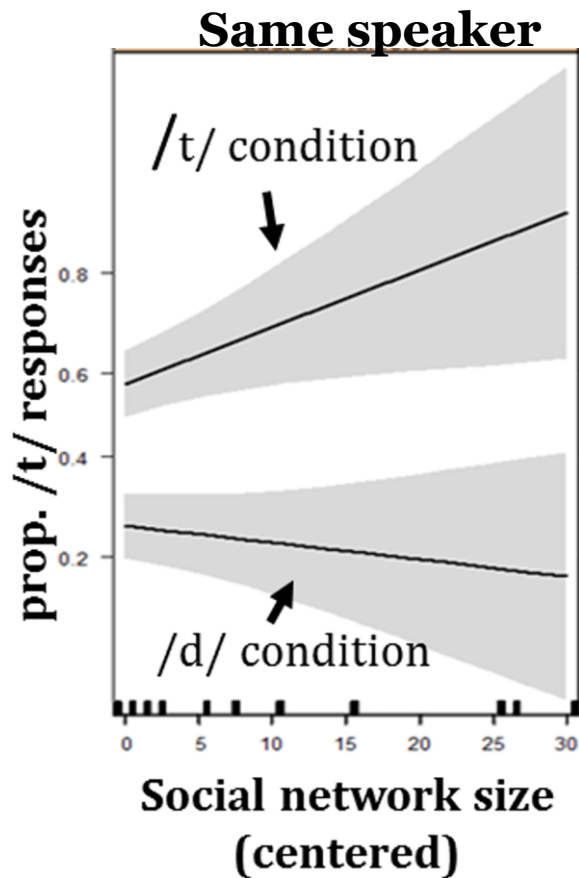
Social network effect



Social network effect



Social network effect

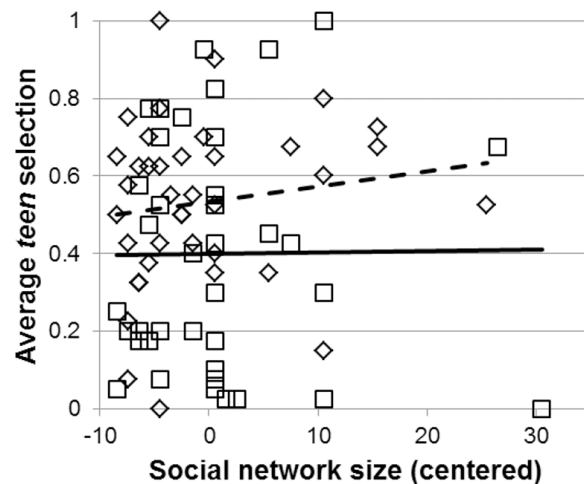


Big social network, little effect of training

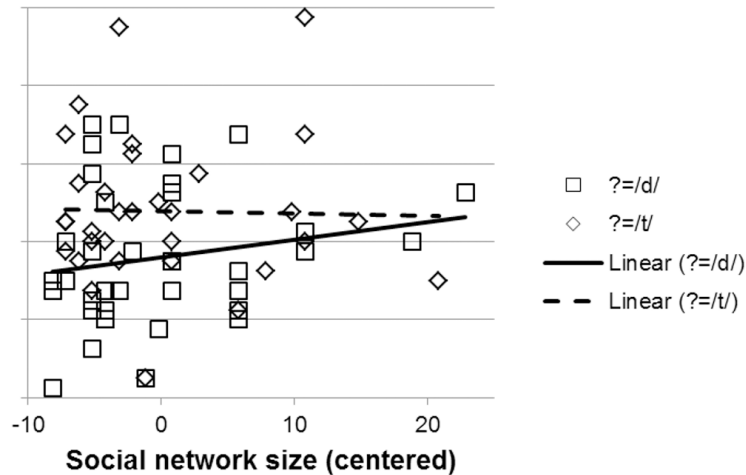
Small social network, big effect of training

Raw data

Same speaker

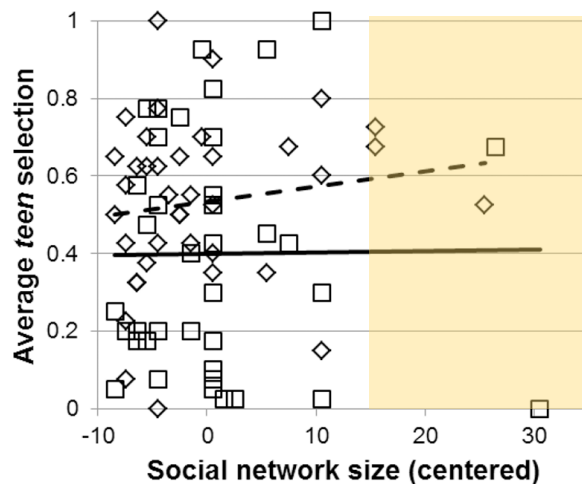


New speaker

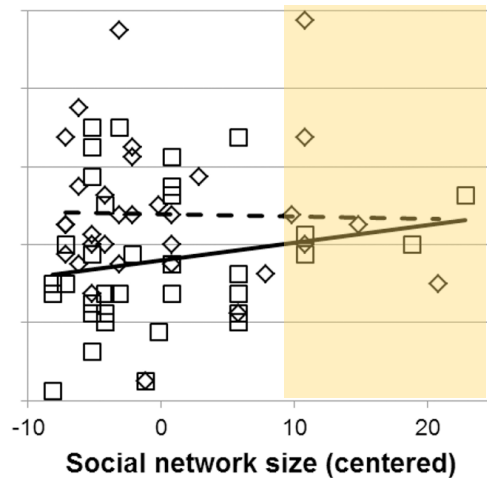


Raw data

Same speaker



New speaker



- $\square$ $?=/d/$
- $\diamond$ $?=/t/$
- Linear ($?=/d/$)
- - Linear ($?=/t/$)

Alternative analyses

From supplementary materials

“1. Analyses that do not exclude the participants whose Social Network Size was 4 and 8 standard deviations from the mean show very different results from those reported in the paper. Comparing results with and without a particular participant or observation is in fact the test for undue influence, often called Cook’s Distance [1]. Analyses whose results depend on including a specific data point or individual are considered unreliable.

2. Analyzing the data with the untrimmed estimates of Social Network Size leads to the same pattern of results as in the analysis reported in the paper, but the triple interaction does not reach conventional level of significance ($\beta=0.13263$, $SE=0.07405$, $z=1.791$, $p=0.073$).”

Lev-Ari's conclusions

Individuals with smaller social networks have more malleable linguistic representations

- More influenced by manipulated-/d/ training
- Potential role for such individuals in propagating linguistic change?
 - For further development and testing of this idea, see [Lev-Ari, 2018](#)

**Time for Q&A/discussion on this
week's reading**

Next up

Thursday lab

- A perceptual learning experiment using Lev-Ari's audio stims (thanks Shiri!) and equivalent visual stims

Next week

- Priming and overspecification, recording participant audio

